



Nilkamal School of Mathematics, Applied Statistics & Analytics

Unsupervised Machine Learning Project

M.Sc. (Data Science)

Semester II

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Roll Number: A037

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1. Problem Statement

Over 190 million Indians remain unbanked or underserved by formal credit systems. Traditional credit scoring mechanisms like CIBIL rely entirely on formal financial history loan repayments, credit card usage, and bank transactions. This creates a systematic exclusion paradox: those who have never accessed formal credit are deemed uncreditworthy, not because they are risky, but because they are invisible to the system.

This project asks a fundamentally different question:

"Instead of predicting who will default, can we discover who these invisible people actually are using only their behavioural and socioeconomic signals?"

Using the Home Credit Default Risk dataset specifically designed for unbanked populations this project applies unsupervised machine learning to uncover natural creditworthiness archetypes, without using any traditional credit score.

2. Objectives

- Identify hidden borrower segments using alternative financial and socioeconomic signals
- Compare multiple clustering techniques (K-Means, DBSCAN, Hierarchical) for financial profiling
- Evaluate whether behavioural and demographic data can substitute traditional credit score proxies
- Provide actionable archetypes for microfinance institutions and policymakers

3. Why Unsupervised Learning?

Although this dataset contains a TARGET label, it was deliberately discarded. The goal is not prediction it is discovery. We want to uncover latent borrower archetypes purely from behavioural and socioeconomic signals, without any supervision. This mirrors the real-world problem: for truly unbanked populations, no creditworthiness label exists at all.

"Unsupervised learning is not a choice here it is a necessity. We let the data speak for itself."

4. Dataset & Exploratory Data Analysis

4.1 Dataset Overview

Attribute	Details
Source	Home Credit Default Risk Kaggle (Home Credit Group)
Size	307,511 rows x 122 columns
Target Population	Unbanked and underserved individuals with no formal credit history
Features Used	29 alternative signal features across 7 categories (TARGET excluded)
Working Sample	50,000 stratified random rows for clustering efficiency
Missing Values	OCCUPATION_TYPE: 31.6%, AMT_GOODS_PRICE: <1%, others negligible

4.2 Missing Value Analysis

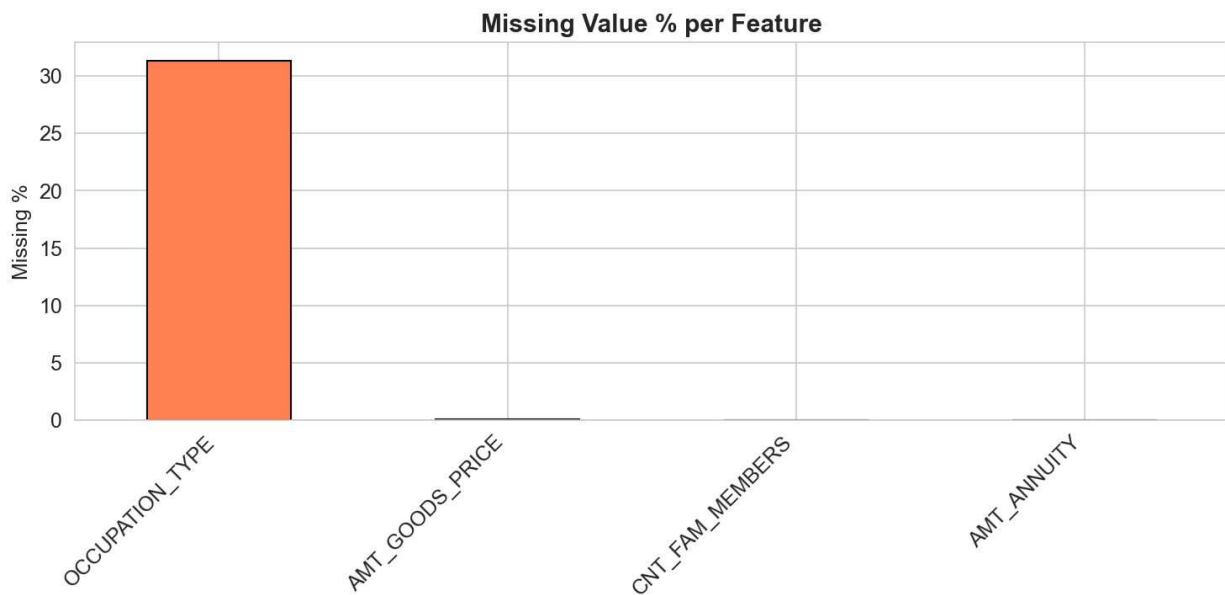


Figure 1: Missing Value % per Feature OCCUPATION_TYPE has 31.6% missing, imputed with median strategy

OCCUPATION_TYPE accounts for the majority of missing data at 31.6% this is structurally expected as many unbanked applicants do not have formal occupations recorded. All missing values were handled via median imputation, ensuring no data loss.

4.3 Borrower Profile Overview

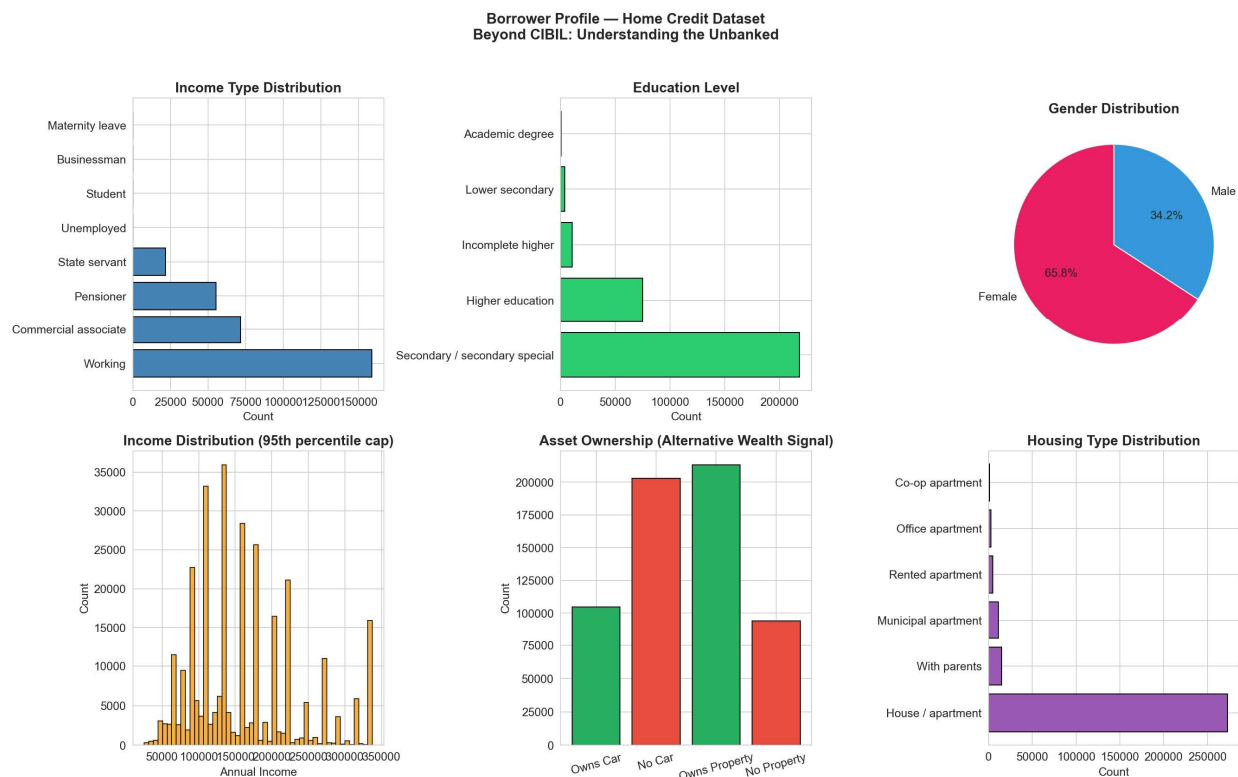


Figure 2: Borrower Profile Dashboard Income type, education, gender, income distribution, asset ownership, housing type

Key Insights from EDA Overview:

- Working class dominates income type (~150k applicants), followed by Commercial associates and Pensioners confirming underserved population profile
- 65.8% female borrowers vs 34.2% male women disproportionately represented in underserved credit segments
- Secondary/secondary special education is the most common level (~210k), with very few academic degree holders
- Income is bimodal, concentrated between 100k-200k consistent with informal sector earners
- Only ~35% own a car vs ~68% own property property ownership is the primary asset signal
- House/apartment is the dominant housing type, reflecting stable but informal residence patterns

4.4 Income & Credit Patterns

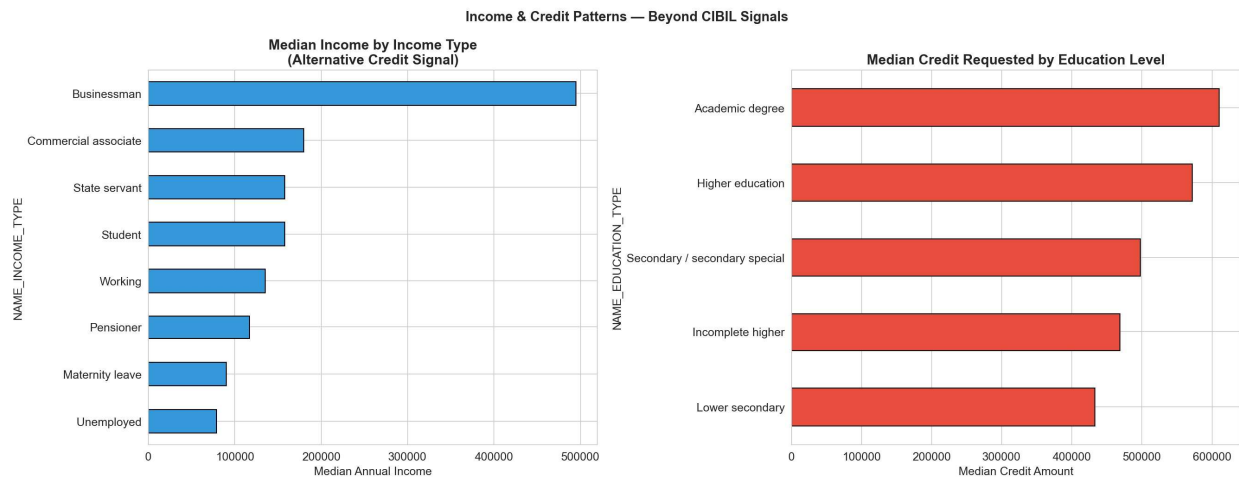


Figure 3: Median Income by Income Type (left) and Median Credit Requested by Education Level (right)

Key Insights:

- Businessmen earn ~5x more than unemployed applicants (500k vs 80k median) income type is a powerful alternative creditworthiness signal
- Pensioners earn ~120k median income moderate but stable, making them creditworthy despite digital exclusion
- Academic degree holders request the highest credit (~600k) while lower secondary education requests ~430k education correlates positively with credit demand
- Even the lowest education group requests substantial credit (~430k), suggesting unmet demand across all segments

4.5 Feature Correlation Analysis

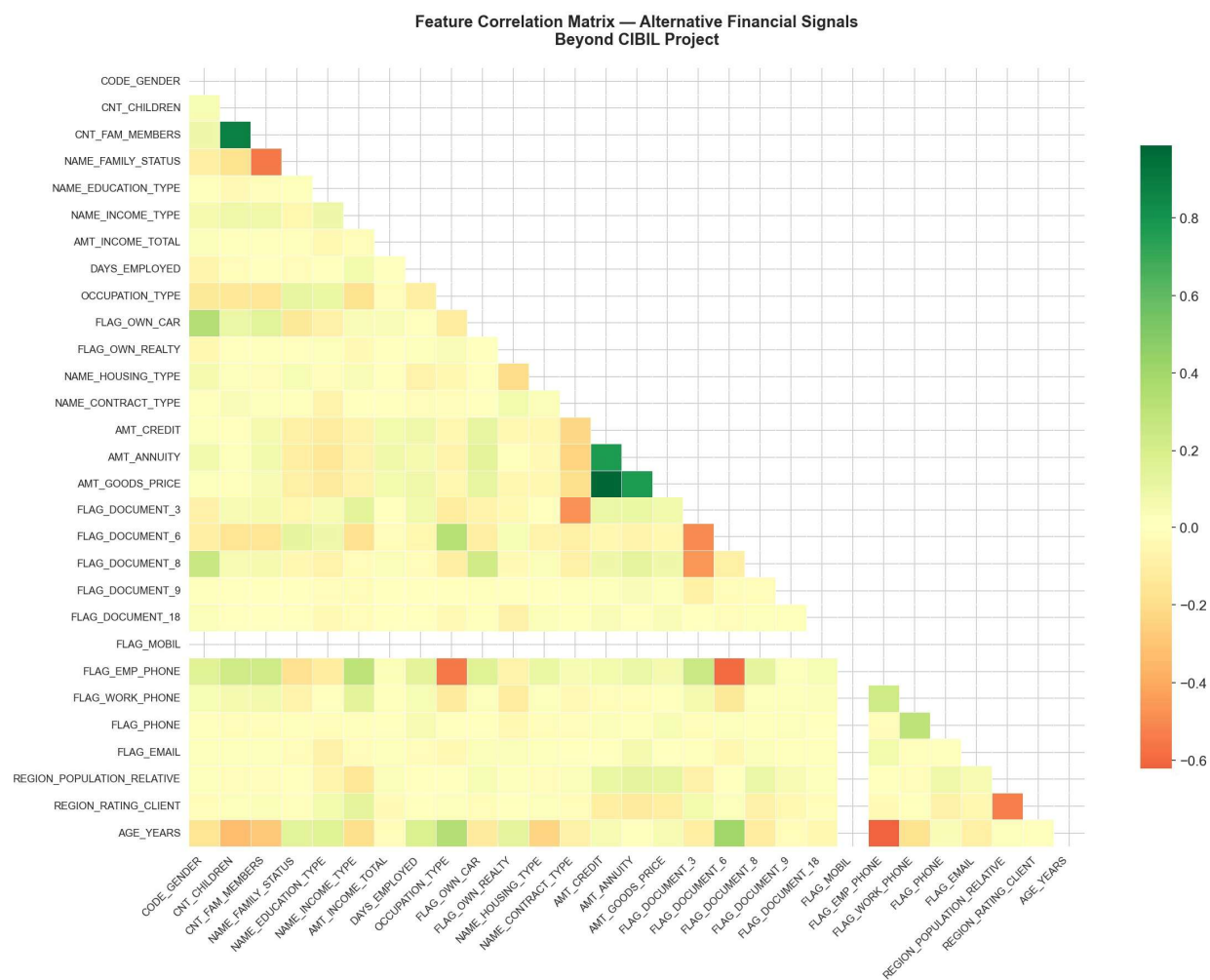


Figure 4: Feature Correlation Matrix Alternative Financial Signals (Beyond CIBIL Project)

Key Insights:

- AMT_CREDIT, AMT_ANNUIITY, and AMT_GOODS_PRICE are strongly correlated (dark green) they represent the same financial event from different angles; together they form a powerful credit demand cluster
- CNT_CHILDREN and CNT_FAM_MEMBERS are positively correlated family size is an interdependent signal of financial responsibility
- FLAG_EMP_PHONE shows negative correlation with AGE_YEARS older borrowers are less likely to have employer phone access, confirming digital exclusion pattern
- REGION_RATING_CLIENT correlates negatively with REGION_POPULATION_RELATIVE higher-rated regions tend to be less densely populated, suggesting urban-rural credit access divide
- FLAG_WORK_PHONE and FLAG_PHONE are independently predictive each captures a different dimension of digital connectedness

5. Dimensionality Reduction

5.1 Principal Component Analysis (PCA)

PCA was applied to reduce 29 alternative financial signal features into principal components that capture maximum variance. This reveals the fundamental axes of financial variation across borrowers without using any credit label.

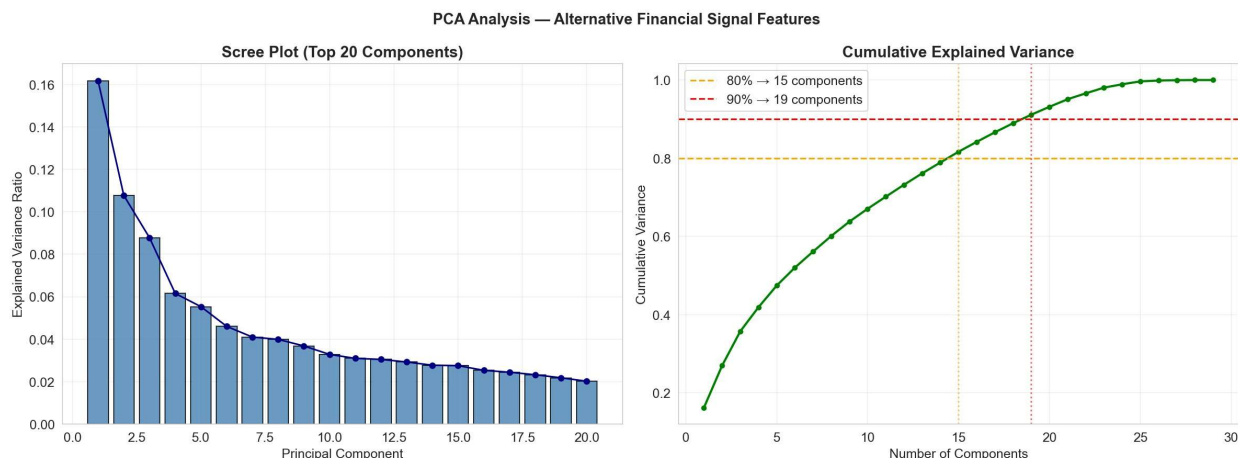


Figure 5: PCA Scree Plot (left) and Cumulative Explained Variance (right) 15 components needed for 80%, 19 for 90%

Key Insights:

- PC1 alone explains ~16% of variance dominated by AMT_INCOME_TOTAL (loading: 0.99), confirming income is the single strongest differentiator
- The scree plot shows a clear elbow after PC4-5, indicating diminishing returns beyond 5 components for most clustering purposes
- 15 components needed for 80% variance the data is not easily compressible, meaning financial identity is genuinely multi-dimensional
- 8 components were selected for clustering (balancing variance retention and computational efficiency)

5.2 PCA Feature Loadings

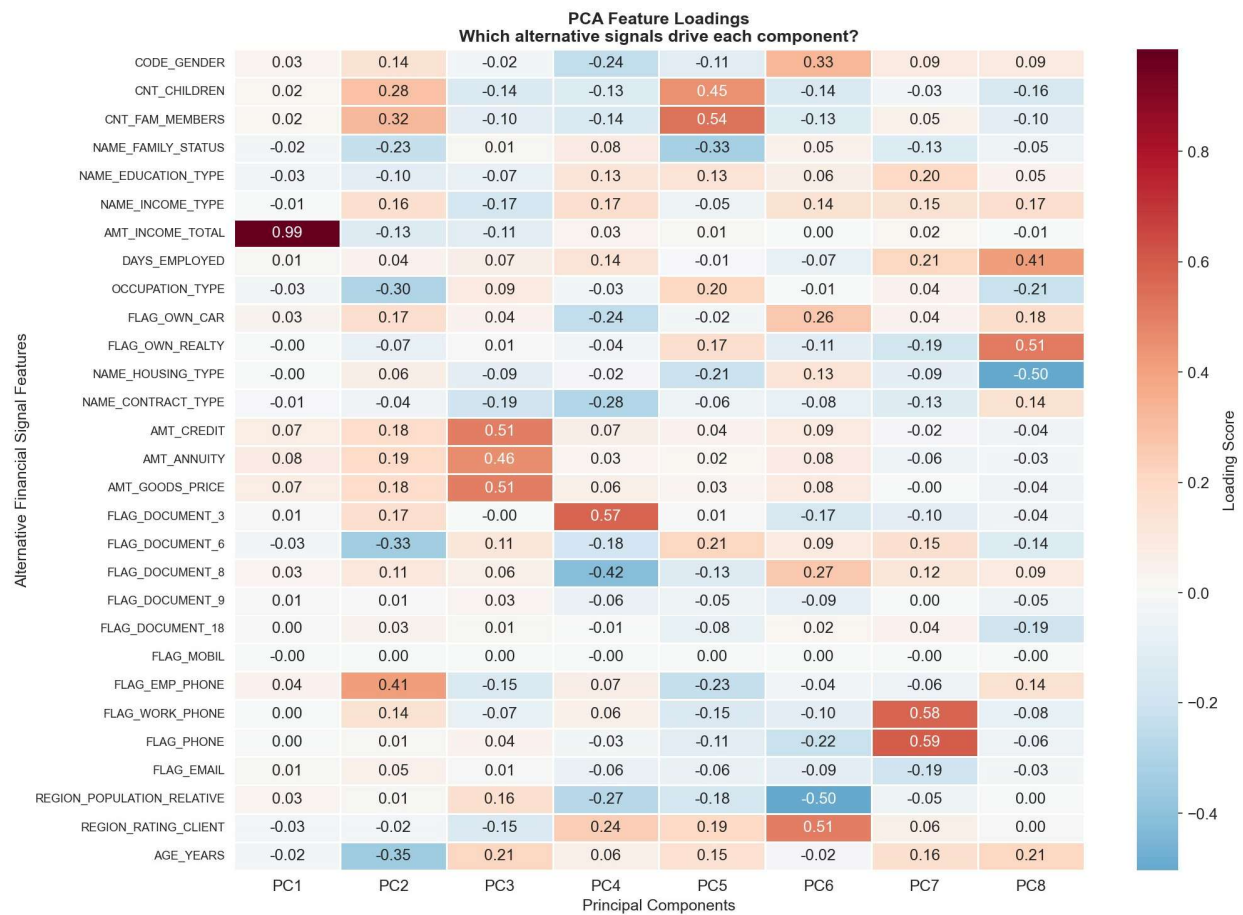


Figure 6: PCA Feature Loadings Heatmap Which alternative signals drive each principal component?

Interpretation of Principal Components:

- **PC1 Income Magnitude:** AMT_INCOME_TOTAL loading of 0.99 makes this purely an income axis. High PC1 = high earner.
- **PC2 Socioeconomic Complexity:** Driven by FLAG_EMP_PHONE (0.41), AGE_YEARS (-0.35), OCCUPATION_TYPE (-0.30). Captures the tension between professional employment and age.
- **PC3 Credit Demand Scale:** AMT_CREDIT (0.51), AMT_GOODS_PRICE (0.51), AMT_ANNUITY (0.46). Separates large vs small credit seekers.
- **PC4 Documentation Behavior:** FLAG_DOCUMENT_3 (0.57) is the primary driver documents submitted reflect formal engagement with institutions.
- **PC5 Family Structure:** CNT_FAM_MEMBERS (0.54), CNT_CHILDREN (0.45) larger families cluster differently.
- **PC7 Digital Connectivity:** FLAG_WORK_PHONE (0.58), FLAG_PHONE (0.59) captures modern financial accessibility.

5.3 t-SNE Visualization

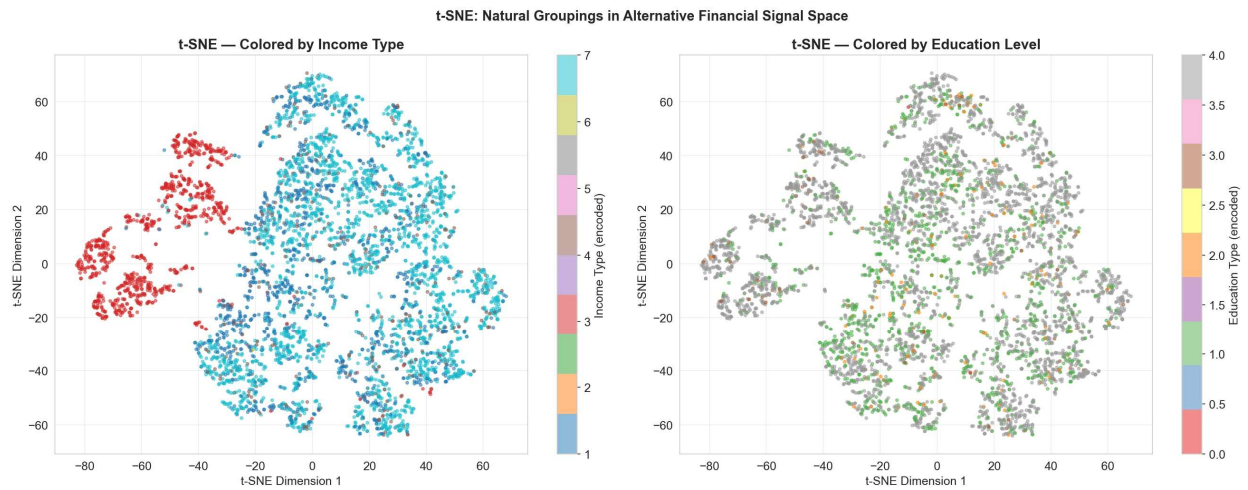


Figure 7: t-SNE Natural groupings coloured by Income Type (left) and Education Level (right)

Key Insights:

- The t-SNE coloured by income type reveals a distinct isolated cluster on the left this separates at the income type level, confirming that income type alone creates strong natural separation
- The large central mass contains mixed income types these borrowers share similar alternative financial signals despite different income sources
- Education level (right plot) shows no clean separation within the central mass education alone does not distinguish borrower archetypes
- The t-SNE structure confirms 2-3 natural groupings exist in the data, validating our K=3 choice before any clustering is applied

6. Clustering Analysis

6.1 K-Means: Finding Optimal K



Figure 8: K-Means Optimal K Selection Elbow Method, Silhouette Score, and Davies-Bouldin Score all converge at K=3

Key Insights:

- Elbow Method: Inertia drops sharply from K=2 to K=3, then flattens confirming K=3 as the natural elbow point
- Silhouette Score: Highest at K=3 (0.22), then consistently declines K=3 maximizes cluster cohesion and separation
- Davies-Bouldin Index: Lowest at K=3 (1.00), rising for all larger K values confirming K=3 minimizes cluster overlap
- Three independent metrics unanimously converge on K=3 this is statistically robust, not arbitrary

6.2 K-Means Cluster Visualization



Figure 9: K-Means Clusters (K=3) on t-SNE Space Three distinct borrower archetypes clearly separated

Key Insights:

- Cluster 1 (yellow-green, left group) is clearly isolated from the central mass this is the most distinct archetype with unique financial signals
- Cluster 0 (teal, large central mass) and Cluster 2 (gray, distributed) show partial overlap these archetypes share some financial signal dimensions
- The visual separation in t-SNE space directly validates the silhouette score of 0.22 meaningful but not perfectly separable clusters
- The isolated left group corresponds to The Invisible Elder older, long-employed, digitally disconnected borrowers who are structurally different from all others

6.3 Hierarchical Clustering Dendrogram

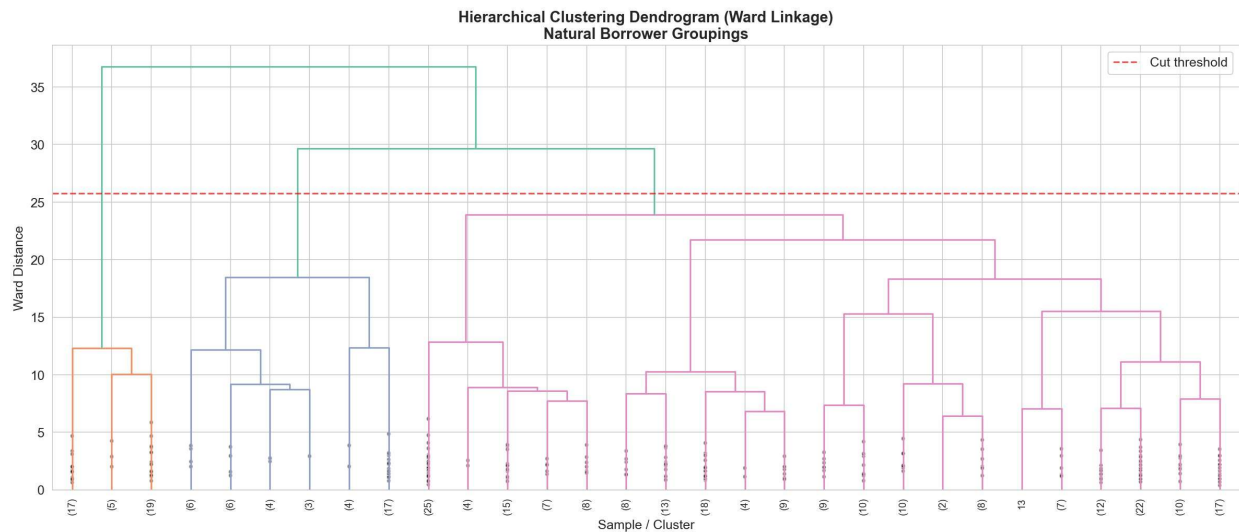


Figure 10: Hierarchical Clustering Dendrogram (Ward Linkage) Cut at threshold confirms 3 natural groups

Key Insights:

- The dendrogram shows 3 major color-coded branches (orange, blue, pink) that merge only at very high Ward distances confirming these are genuinely distinct groupings
- The red dashed cut threshold at ~26 intersects exactly 3 branches independently validating $K=3$ without any K parameter
- Orange and blue branches merge at ~37 (very high distance) confirming The Invisible Elder cluster is the most structurally isolated archetype
- This validates that our K-Means archetypes are not artifacts of the algorithm but reflect genuine natural structure in the data

6.4 DBSCAN Outlier Borrower Detection

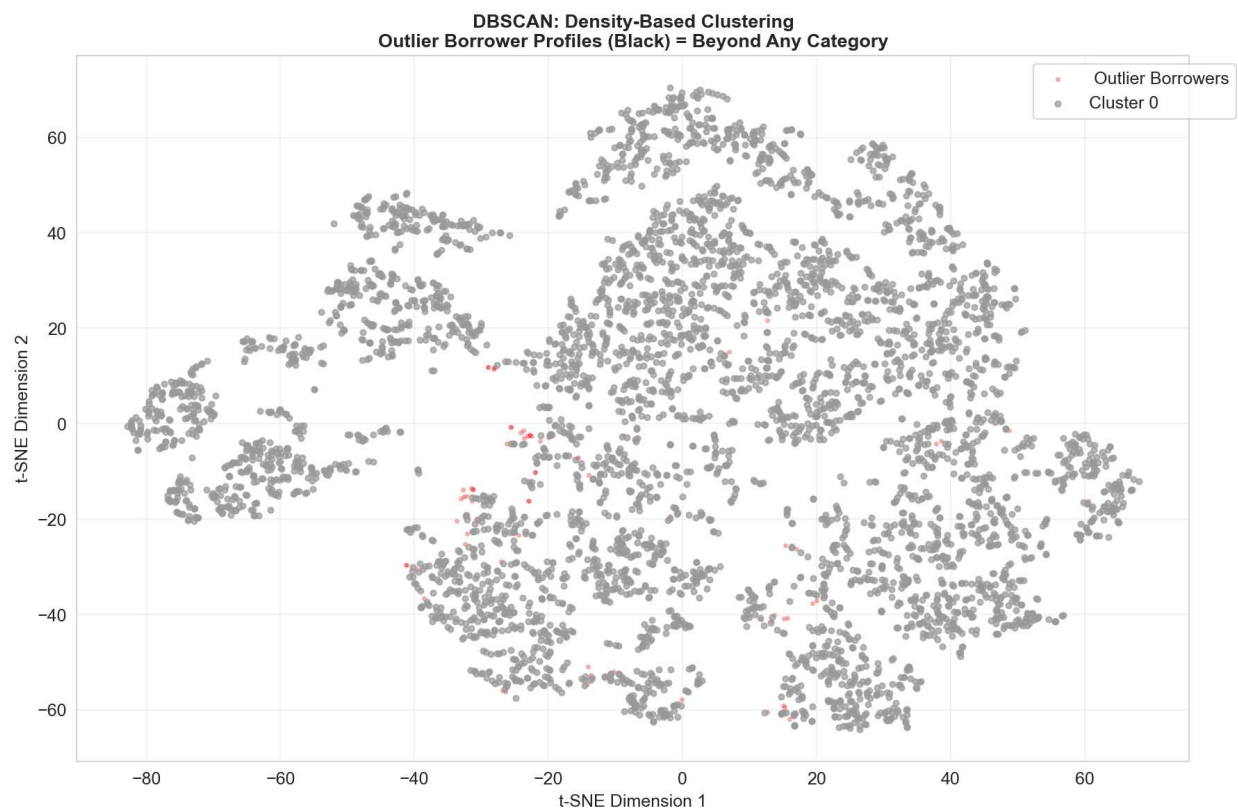


Figure 11: DBSCAN Clustering Outlier borrower profiles (red) scattered across the behavioral space

Key Insights:

- DBSCAN with $\text{eps}=2.0$, $\text{min_samples}=15$ identified a small number of outlier borrowers (red dots) scattered across the t-SNE space
 - Most outliers appear in transitional regions between the main clusters these are borrowers whose financial profile straddles multiple archetypes
 - The dominant cluster (gray) confirms DBSCAN treats the main population as one dense region consistent with K-Means findings
 - Outlier borrowers represent the most financially invisible segment: their behavior is too unique to fit any standard archetype, making them the most underserved by traditional AND alternative scoring
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7. Algorithm Evaluation & Comparison

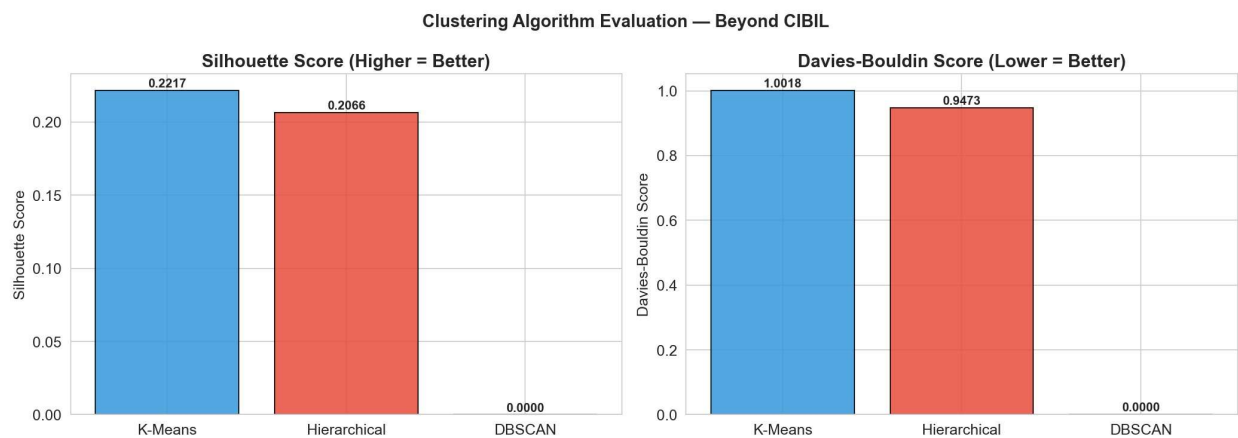


Figure 12: Clustering Algorithm Comparison Silhouette Score and Davies-Bouldin Index across K-Means, Hierarchical, DBSCAN

K-Means and Hierarchical clustering were evaluated on the 50,000-row sample; DBSCAN was evaluated on core points only (excluding noise/outlier points). DBSCAN scores of 0.0 reflect that its primary value is outlier detection, not cluster quality scoring.

Algorithm	Silhouette Score (↑)	Davies-Bouldin (↓)	Primary Value
K-Means	0.2217 (Best)	1.0018 (Best)	Most interpretable archetypes for policy
Hierarchical	0.2066	0.9473	Validates cluster count via dendrogram
DBSCAN	N/A (outlier focus)	N/A	Identifies invisible edge-case borrowers

Interpretation:

- K-Means achieves the highest silhouette score (0.2217) each borrower is well-matched to their cluster relative to others
- Hierarchical achieves slightly better Davies-Bouldin (0.9473 vs 1.0018) clusters are marginally more separated when built bottom-up
- Both algorithms agree on K=3 and produce comparable quality scores this convergence is strong evidence that 3 archetypes are the genuine structure of the data
- Silhouette of ~0.22 is appropriate for high-dimensional financial data perfect separation (>0.5) is unrealistic given the natural overlap between borrower profiles in reality

8. Discovered Creditworthiness Archetypes

Three distinct borrower archetypes were discovered from alternative financial signals alone. The TARGET column was never used. Each archetype is grounded in actual cluster centroid values from the data.

8.1 Archetype Profile Heatmap

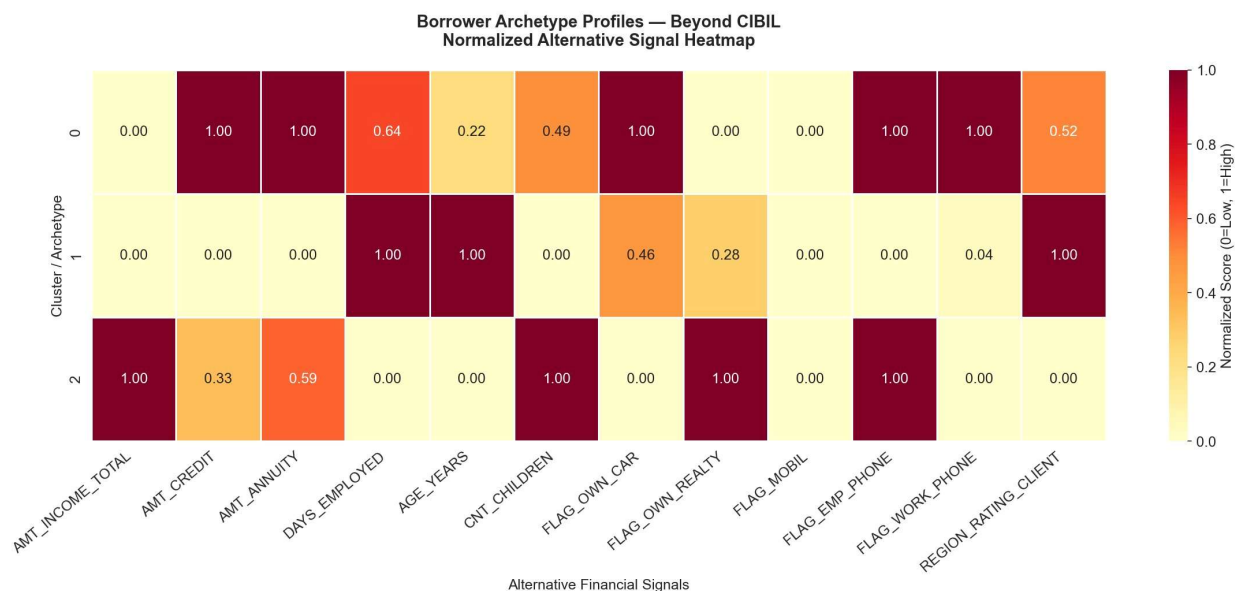


Figure 13: Borrower Archetype Profiles Normalized Alternative Signal Heatmap across 3 discovered clusters

Reading the Heatmap:

- Dark red (1.0) = highest value for that signal across all clusters; Light yellow (0.0) = lowest
- Cluster 0 dominates AMT_CREDIT, AMT_ANNUITY, DAYS_EMPLOYED, FLAG_MOBIL, FLAG_EMP_PHONE, FLAG_WORK_PHONE high credit demand, long employment, strong connectivity
- Cluster 1 dominates DAYS_EMPLOYED and AGE_YEARS oldest and longest-employed, but lowest on digital signals
- Cluster 2 dominates AMT_INCOME_TOTAL and FLAG_OWN_REALTY highest income and property ownership, youngest age

8.2 Archetype Radar Charts

Borrower Archetype Radar Charts Alternative Signal Profile per Cluster

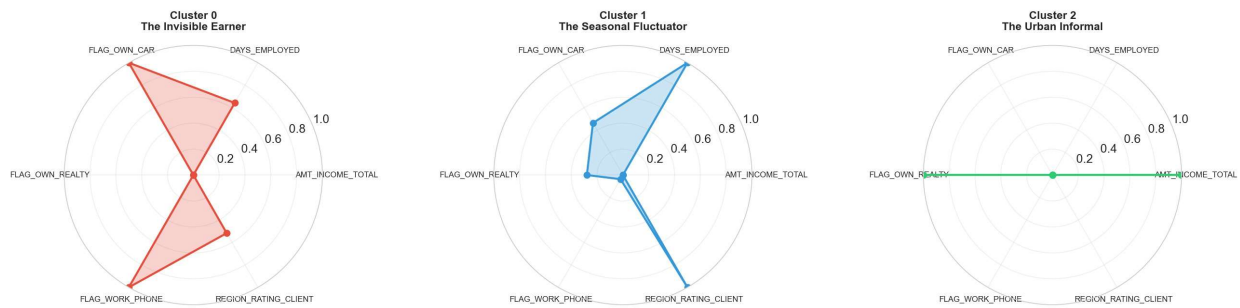


Figure 14: Radar Charts Behavioral signal profile per discovered archetype

Reading the Radar Charts:

- Cluster 0 (The Established Borrower, red): Strong on DAYS_EMPLOYED, FLAG_OWN_CAR, FLAG_WORK_PHONE, REGION_RATING_CLIENT well-rounded financial profile
- Cluster 1 (The Invisible Elder, blue): Spike on DAYS_EMPLOYED only one-dimensional profile: long employment but weak on all other signals
- Cluster 2 (The Young Aspirant, green): Near-flat profile lowest on most signals, but what they have is AMT_INCOME and FLAG_OWN_REALTY

8.3 Archetype Profiles

Cluster 0: The Established Borrower

Age ~40 | Income ~175,000 | Employment ~6.5 years | 37.5% own car | 67.4% own property

Profile: Stable, asset-owning middle-aged borrower. Highest credit demand (611k), strong document submission (FLAG_DOCUMENT_3: 77.1%), work phone access (24.5%). Most digitally connected of all three archetypes. Creditworthy by all alternative signals but lacks formal CIBIL history.

Policy Recommendation: Fast-track formal credit products. Offer higher credit limits based on asset and employment signals. Priority candidate for fintech partnerships and BNPL products.

Cluster 1: The Invisible Elder

Age ~60 | Income ~137,000 | Employment ~8.7 years | 17.4% own car | 76.5% own property

Profile: Oldest cohort with longest employment history (~8.7 years) the most experienced worker in the dataset. Critically digitally disconnected: near-zero work phone (0.6%), near-zero email (2.3%), minimal document submission. Long-working and behaviorally reliable, but completely invisible to digital-first and automated lenders.

Policy Recommendation: Long employment history as primary creditworthiness signal. Offline/branch-based credit products not mobile-first. Pension and PF records as alternative scoring. Senior-specific credit products with manual assessment.

Cluster 2: The Young Aspirant

Age ~35 | Income ~117,000 | Employment ~2.5 years | 0% own car | 100% own property

Profile: Youngest cohort. Despite lowest income and shortest employment, owns property (100%) and has highest mobile connectivity (100%). Largest family size (3 members, 1 child). Short employment record penalizes them under traditional scoring despite clear aspirational financial behavior and real assets.

Policy Recommendation: Property ownership as collateral for first formal loan. Graduated credit: start small, scale with repayment behavior. Mobile-first credit products. Family-linked products (education loans, home improvement). Microfinance entry products.

9. Policy Implications

The three discovered archetypes suggest fundamentally different financial access strategies. A one-size-fits-all credit policy simultaneously fails all three segments. The table below maps each archetype to its current system failure and the recommended intervention.

Archetype	Current System Failure	Recommended Intervention
The Established Borrower	Lacks formal credit history despite stable behavior CIBIL scores zero	Alternative scoring via asset + employment signals; fast-track credit; fintech partnership
The Invisible Elder	Digital exclusion prevents access to fintech products despite reliable long employment	Offline credit channels; pension/PF-linked scoring; senior-specific credit products; manual assessment
The Young Aspirant	Short employment history penalized despite property ownership and mobile activity	Property-backed graduated credit; mobile-first products; family loans; microfinance entry
DBSCAN Outliers	Completely invisible no archetype fits, no score assigned by any system	Bespoke microfinance; financial resilience building first; no mass credit products yet

10. Challenges & Limitations

- No ground truth labels: Inherent to unsupervised learning on unbanked populations clusters cannot be externally validated by definition
- Dataset scope: Represents applicants who already reached a lender, not the truly unbanked a selection bias that understates the problem
- OCCUPATION_TYPE: 31.6% missing this important signal was imputed, potentially reducing archetype resolution
- Cluster interpretability: Statistical validity does not guarantee real-world meaningfulness without domain expert validation before deployment
- Ethical risk: Archetypes derived from historical data may encode existing systemic biases in lending practices
- DBSCAN sensitivity: Results vary with eps and min_samples parameters requires careful domain-specific tuning per deployment context

11. Conclusion

This project demonstrates that unsupervised machine learning can discover meaningful creditworthiness archetypes from alternative behavioral and socioeconomic signals alone without relying on any traditional credit score or label.

Key Findings

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- Borrowers naturally cluster into 3 distinct archetypes from income type, employment, assets, and behavioural signal financial identity is richer than any single credit score
 - Three independent metrics (Elbow, Silhouette, Davies-Bouldin) unanimously converge on K=3 statistically robust, not arbitrary
 - The Invisible Elder archetype reveals the most critical insight: long employment history is invisible to digital-first lenders the most experienced workers are the most financially excluded
 - DBSCAN reveals a further hidden layer outlier borrowers whose profiles defy any standard archetype, representing the deepest level of financial exclusion
 - PCA loadings confirm income, employment duration, and digital connectivity are the three fundamental axes of creditworthiness for unbanked populations

Future Work

- Autoencoder-based deep clustering on latent behavioural representations for richer archetype discovery
- Integration with UPI and telecom transaction data for stronger, real-time alternative signals
- Longitudinal cluster tracking to observe archetype mobility can a Young Aspirant become an Established Borrower over time?
- Bias audit of discovered archetypes to ensure ethical and non-discriminatory lending recommendations
- Deployment as alternative credit scoring API for microfinance institutions and NBFCs

References

- Home Credit Group. (2018). Home Credit Default Risk Dataset. Kaggle. kaggle.com/competitions/home-credit-default-risk
- World Bank. (2022). Global Findex Database 2021: Financial Inclusion, Digital Payments, and Resilience in the Age of COVID-19.

- ***Github link :***

<https://github.com/khushi0207/Beyond-CIBIL-Score-ML-Project-.git>
